

# Extraction and Identification of Medications from EHR Notes with LLMs: A Tool and a Case Study

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**Abstract.** Extracting medication data from Electronic Health Record (EHR) notes plays a critical role in enhancing healthcare Process Mining (PM) by enabling detailed analysis of patient treatment trajectories, medication adherence, and clinical decision-making. However, accurate extraction of information in general, and medication in particular, from unstructured clinical texts remains challenging, particularly when working with non-English datasets and limited Natural Language Processing (NLP) resources. To address these challenges and streamline the process, accessible and efficient methods are needed to enable faster in-house extraction of medication information while complying with privacy regulations. In this paper, we present a case study involving patients with Chronic Obstructive Pulmonary Disease (COPD) from a Spanish-language EHR dataset. We explore the use of a prompt-based approach leveraging an off-the-shelf Large Language Model (LLM) to extract medication names from free-text EHR notes. We also present a tool that enables interactive identification and validation of extracted medication based on SNOMED-CT terminology. The extracted data was successfully integrated into a PM model, demonstrating the practical feasibility and utility of this method for improving the agility and depth of healthcare data analysis workflows.

**Keywords:** LLMs · Medication Extraction · Electronic Health Records · Process Mining

## 1 Introduction

Electronic Health Records (EHRs) are frequently recorded as unstructured, free-text documents, making them both a rich source of clinical information and a significant challenge for automated analysis. Extracting structured data—such as medication mentions, diagnoses, or treatment actions—is critical for enabling advanced applications in healthcare, including clinical decision-making, patient monitoring, and Process Mining (PM). PM is a methodology for uncovering patient circuits and treatment trajectories, enabling healthcare providers to better

understand complex processes based on real-world data, identify non-standard patients by analyzing infrequent behavior, and ultimately evolve clinical protocols iteratively using these results [1]. However, the unstructured nature of EHRs often creates barriers to integrating such data into PM workflows [3].

Existing Natural Language Processing (NLP) techniques have been explored to extract structured information from clinical text [5], with many studies using rule-based systems, machine learning models, and fine-tuned neural networks [6], as well as the creation of specific NLP tooling [4]. While these methods are powerful, they often require extensive manual annotation, domain-specific expertise, and annotated datasets, making them inaccessible to many healthcare IT teams. These barriers highlight the need for accessible, in-house solutions that comply with data protection standards while enabling efficient information extraction.

Recent advancements in large language models (LLMs), such as GPT-based architectures, offer a promising alternative. LLMs have demonstrated strong performance across a wide range of NLP tasks, including medical information extraction, without requiring fine-tuning or large annotated datasets [7]. By leveraging natural-language prompts, these models could allow practitioners with limited AI expertise to extract structured information directly from unstructured clinical text. However, the application of LLMs to non-English EHRs poses unique challenges, such as limited availability of high-quality annotated datasets and linguistic variations that may affect model performance [8].

In this paper, we present a practical case study using a general-purpose LLM (Llama 3 [11]) to extract medication names from Spanish-language EHR notes and integrate them into PM models. We explored various prompt configurations to address challenges such as output format, language, and scope, ultimately evaluating the performance of a selected approach. We also present a tool that enables interactive identification and validation of extracted medication using SNOMED-CT terminology. Our approach demonstrates how unstructured clinical data can be transformed into structured formats suitable for PM, leveraging LLMs to bypass the need for resource-intensive annotation processes and domain-specific model fine-tuning.

This work represents a first step, in a real context, for leveraging LLMs to bridge the gap between unstructured clinical text and PM workflows, offering insights that may inform future developments in this area.

## 2 Background

### 2.1 Process Mining in Healthcare

Fernández-Llatas et al. define PM as a “*Syntactic Data Mining technique that supports the domain experts in the proper understanding of complex processes in a Comprehensive, Objective way.*” [1]. Applying this technique typically involves using structured data sources such as ICD codes, timestamps, and medication administration records to construct event logs. However, as Fox et al. [3] demonstrate in their development of a care pathway data quality framework (CP-DQF),

addressing data quality issues is critical for ensuring reliable EHR-based research and PM outcomes. Their work highlights the importance of data quality management for structured data, as illustrated through a case study on dental care pathways. Yet, despite its comprehensive approach, their framework does not address the challenges associated with unstructured free-text data, such as clinical notes, which often contain rich descriptions of patient conditions, treatments, and care decisions. Munoz-Gama et al. [2] identify data quality as a distinguishing challenge of PM in healthcare (challenge C6). This is particularly true for free-text clinical notes, which are often manually recorded by clinicians or administrative staff.

The reliance on structured data in current PM frameworks overlooks the wealth of information embedded in unstructured clinical notes. These notes, while valuable, are often characterized by issues such as inconsistent terminology and incomplete records, making their integration into PM applications challenging. Our study bridges this gap by leveraging LLMs to extract and structure medication data from unstructured EHRs, enabling its integration into PM workflows and enhancing the accessibility and usability of this previously untapped information source.

## 2.2 Natural Language Processing for Clinical Text

NLP has been extensively studied for extracting structured information from clinical text [4, 5]. Traditional NLP methods, such as rule-based systems and fine-tuned neural networks, have proven effective in certain contexts but often require substantial resources, including annotated datasets, computational power, and domain expertise. For example, Hangyu et al. [6] propose a five-step methodology based on embedding techniques and complex deep learning systems, such as convolutional neural networks (CNN). While effective, these methods can be impractical for resource-constrained healthcare IT teams due to their high implementation and maintenance costs.

These challenges are further magnified in non-English languages, such as Spanish, as noted by Névéol et al. [8]. Adapting NLP methods to Spanish-language EHRs introduces unique difficulties, including linguistic variability, abbreviations, mixed medical terminology, and a lack of annotated corpora. For instance, abbreviations and mixed terminologies frequently lead to ambiguities that traditional NLP systems struggle to resolve without substantial contextual understanding. These limitations complicate efforts to extract structured information from clinical text in Spanish, highlighting the need for more adaptable and resource-efficient approaches.

## 2.3 LLMs in Healthcare

In recent years, LLMs have demonstrated their potential for NLP tasks in healthcare, including information extraction [7]. Unlike traditional methods, LLMs can perform tasks without fine-tuning, relying instead on natural-language prompts to guide their outputs. This makes them particularly appealing for practitioners

without AI expertise. However, most studies focus on English-language data, with limited exploration of their applicability to non-English clinical text. Our study contributes to this emerging area by evaluating the performance of LLMs on Spanish-language EHRs.

### 3 Materials and methods

#### 3.1 Technical Resources

The LLM used for medication extraction was **Llama 3** [11] (specifically, the 8-billion-parameter model, “*llama3:8b*”). It was chosen because it is a powerful and widely recognized open-source model that supports self-hosting, which was essential for this experiment.

To host the model locally, we used **Ollama**, a lightweight and flexible server designed for hosting foundation models. Ollama provides a RESTful API interface, allowing seamless interaction with the model and enabling the configuration of parameters such as temperature, output format, and system instructions. This setup ensured efficient and secure execution of the extraction tasks.

The ability to self-host the model was critical due to the sensitive nature of clinical patient data. Although the dataset did not include directly identifiable information (such as names or government identifiers), it contained free-text notes that could unintentionally expose sensitive information if proper de-identification processes were not followed. This risk, resulting from potential human errors during data preparation, made it imperative to avoid sending any data to third-party systems.

By hosting the model locally with Ollama, we ensured that all data processing remained within a secure environment, mitigating privacy risks while maintaining full control over the computational infrastructure.

#### 3.2 Clinical Context and Dataset

This study focused on Chronic Obstructive Pulmonary Disease (COPD) [10], a prevalent respiratory condition characterized by persistent airflow limitation and chronic symptoms. Accurate identification of medications administered during emergency visits is essential for understanding COPD management and treatment trajectories.

The dataset used was sourced directly from a real-world hospital information system in Spain and consisted of Spanish-language EHR data structured across multiple tables. The primary table contained emergency care discharge **reports**, organized by field codes.

We disregarded fields containing doctor recommendations due to their nuanced and ambiguous contexts. These fields often include additional contextual information or advice, which makes them more challenging for an LLM to interpret accurately. For example, an LLM could inadvertently extract medications the doctor explicitly advised against, given its limited context-awareness. Furthermore, recommendations may not always reflect actual treatments received,

as they could be ignored by patients or subsequent physicians based on external variables not captured in our dataset.

Instead, we focused exclusively on field code *100*, labeled as “Treatment administered in emergency care”, as it provided the most reliable indication of medications that were actually administered during the emergency visit. This resulted in a total of **842 reports** for analysis. A small sample of the EHR notes found in these reports can be seen in Table 1.

**Table 1.** Sample of EHR notes used in this work (treatment administered in emergency care)

| timestamp           | note                                                         |
|---------------------|--------------------------------------------------------------|
| 22/05/2023 20:46:18 | Ventolin + Atrovent + Budesonida x2 Urbason 60mg             |
| 09/04/2023 22:50:28 | - DA - Hidrocortisona 200 mg IV - Metilprednisolona 60 mg IV |
| 06/06/2023 10:05:52 | Paracetamol 1 gramo IV, Atrovent/Salbutamol (aerosoles).     |

Note that we focused on medications because they are central to the management of COPD exacerbations and reflect important clinical decisions. This is related to our final objective, which is the application of PM to infer the different combinations of drugs administered to patients according to their characteristics. Other event types were also considered (mainly laboratory tests and hospital admissions), but have been omitted from this work for the sake of simplicity. However, events on pharmacy dispensing were not included in the dataset.

## 4 Methodology

This section outlines the process followed to extract medication data from EHR notes for integration into our PM model, focusing specifically on COPD-related discharge reports. To ensure the analysis was limited to relevant cases, discharge reports were filtered using the International Classification of Diseases (ICD). The methodology is divided into three main phases: extraction of medications using an LLM, identification of relevant SNOMED CT [12] terms, and validation of the results. Custom tools were developed to automate and streamline these processes, as described below.

### 4.1 Extraction of Medications using an LLM

Medication extraction was performed using a Large Language Model (LLM), guided by a carefully designed system prompt. To identify the most effective prompt, we systematically tested several variations, as shown in Table 2. These variations explored differences in the following aspects:

- **Language:** Comparing prompts written in English and Spanish.
- Output **format:** Text (one line per medication, all caps to normalize text) or JSON (structured format, returning medication names as-is).

- **Style:** Single-paragraph instructions versus separating each part of the prompt into a bulleted list.
- Amount of **context:** Extracting all medications or restricting extraction to those related to COPD.
- **Misspellings:** Including instructions to correct misspelled medication names or ignoring potential misspellings.

**Table 2.** System prompts tested for medication extraction

| ID | Prompt                                                                                                                                                                                                                                                                                                                                                                                                              | Comments                                                                 |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1  | You are meant to parse any medical data sent to you in Spanish, <b>ONLY</b> returning the names of any medication you find, in all caps, having each medication in a different line. No other characters should be returned.                                                                                                                                                                                        | Simple prompt in English.                                                |
| 2  | You are meant to parse any medical data sent to you in Spanish.<br>Follow <b>STRICTLY</b> these instructions by order priority:<br>- <b>ONLY</b> return the names of any medication you find <b>AS IS</b> , don't say anything more.<br>- If the text is blank, don't say anything, just send a blank message.<br>- The JSON format should be: { "Medicamentos": [ ] }                                              | Prompt in English with JSON output formatting.                           |
| 3  | Tu objetivo es analizar cualquier información médica que te envíe en Español, devolviendo <b>SOLO</b> los nombres de los medicamentos que encuentres, todo en mayúsculas, con cada medicamento en una línea distinta. Ningún otro carácter debería ser devuelto.                                                                                                                                                    | Simple prompt in Spanish.                                                |
| 4  | You are meant to parse any medical data sent to you in Spanish, <b>ONLY</b> returning the names of any medication related to COPD (chronic obstructive pulmonary disease) you find, in all caps, having each medication in a different line. No other characters should be returned.                                                                                                                                | COPD-specific prompt in English.                                         |
| 5  | You are meant to parse any medical data sent to you in Spanish.<br>Follow <b>STRICTLY</b> these instructions by order priority:<br>- <b>ONLY</b> return the names of any medication you find.<br>- If you find medication names with misspellings, correct them before including.<br>- If the text is blank, don't say anything, just send a blank message.<br>- The JSON format should be: { "Medicamentos": [ ] } | Prompt in English with JSON output and spelling correction instructions. |

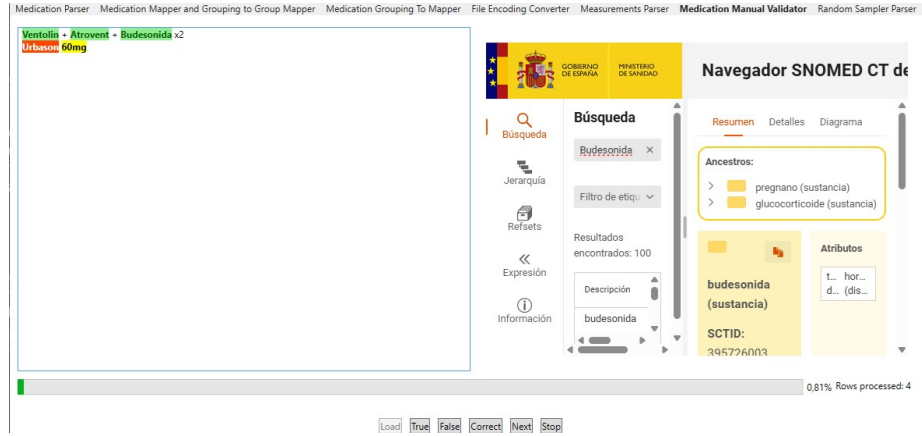
During preliminary testing, a key challenge was observed: the LLM occasionally altered medication names. For example, the brand name “Seguril” was sometimes extracted as “Seguurl” or “Seguiril.” This issue was more common when prompts included additional tasks, such as correcting spelling errors, likely increasing model uncertainty. To mitigate this, we selected a prompt that avoided such tasks (ID 2) and set the temperature parameter to 0, ensuring deterministic outputs.

The “Medication Parser”, a custom tool developed for this study, was used to interact with the LLM, manage API calls, and save the extracted data in a CSV format for downstream processing. An example of the API call is shown below:

```
curl http://localhost:11434/api/generate -d '{
  "model": "llama3",
  "system": "Follow STRICTLY these instructions (...)",
  "prompt": "Ventolin + Atrovent + Budesonida x2\nUrbason 60
mg",
  "options": { \"temperature\": 0 },
  "stream": false
},'
```

## 4.2 Identification of Relevant SNOMED CT Terms

To transform the extracted medications into standardized terminology, the “Medication Manual Validator”, a custom tool developed for this study, was used. This tool, shown in Figure 1, presents the original text on the left and a SNOMED CT search interface on the right, allowing for efficient term identification. Clicking on a medication in the text automatically triggers a SNOMED CT search in the web view, simplifying the process.



**Fig. 1.** Tool developed for this study that allows identification and validation of relevant SNOMED CT terms from the extracted medications. In the example: green marks TPs, red FPs, and yellow FNs

The tool also leverages the Levenshtein distance algorithm [13] to manage discrepancies between extracted medications and the original text. Specifically, it calculates the similarity between strings to:

- Match extracted medications back to the original text, accounting for alterations introduced by the LLM (e.g., “Seguril” extracted as “Seguiril”).
- Enable iterative corrections, where previous manual adjustments made by the user (e.g., harmonizing branded names) are automatically applied to other similar cases, when confirmed.

These features ensure accurate identification and mapping of medications to SNOMED CT terms, while minimizing manual effort.

### 4.3 Analysis and Validation of Results

An expert in the field manually reviewed the extracted medications to ensure correctness. This process involved correcting misspellings, harmonizing commercial names with their generic equivalents, and classifying extracted medications as True Positives (TPs), False Positives (FPs), or False Negatives (FNs).

The “Medication Manual Validator” tool facilitated this validation process with features such as color coding (green for TPs, red for FPs, yellow for FNs, and blue for altered medication names) and underlining for corrected medication names, indicating manual adjustments.

The validation process is further enhanced by the Levenshtein algorithm, which helps reconcile LLM-extracted names with the original text and ensures consistency in iterative corrections. The final output, saved in a CSV format, includes fields for corrected names, classifications (e.g., TP, FP, FN, or TP\* for altered names), and other relevant metadata.

### 4.4 Custom Tools for Medication Parsing

To automate and streamline the workflow, we developed a desktop application that bundles the tools described above. These tools, implemented in C#, are publicly available at <https://github.com/sabienUPV/MineguideEPOCParser>. They include:

- **Medication Parser:** Automates the extraction of medications from clinical notes and outputs results in a CSV format.
- **Medication Manual Validator:** Facilitates both the identification of SNOMED CT terms and the validation of extracted medications in a single iterative process, with features such as automated corrections, interactive validation, and visual cues.

## 5 Results

The medication extraction process identified a total of **3739** individual **instances** of **medication administered** across all patients during emergency visits.

In these events, **122 unique medications** were identified. The most frequently administered medications were *ipratropium bromide* (681 instances) and *salbutamol* (649 instances), two commonly used bronchodilators for COPD patients.



**Table 3.** LLM medication extraction performance based on a sample of 100 notes

| TP  | FP | FN | Precision | Recall | F1-Score |
|-----|----|----|-----------|--------|----------|
| 443 | 10 | 1  | 0.978     | 0.998  | 0.9877   |

### 5.1 Evaluation of LLM performance

To assess the accuracy of the extraction process, we conducted a detailed evaluation on a random sample of 100 reports. The results are presented in Table 3.

To evaluate the performance of the LLM for medication extraction, we applied a consistent methodology that accounted for both combined and separate matches. The evaluation criteria, or ground truth, is decided by using the “Medication Manual Validator” tool as described in section 4.3. Specifically, if the ground truth listed a combination of medications (e.g., “salbutamol+ipratropio”) and the LLM extracted them separately (e.g., “salbutamol” and “ipratropio”), each individual match was counted as a True Positive (TP). In such cases, the combined entity was effectively treated as two distinct TPs. Conversely, if the LLM extracted a combination while the ground truth listed the medications separately, the combined match was treated as a single TP, reflecting the number of matches from the LLM’s perspective, while acknowledging the additional work needed to separate these medications after the extraction process. This approach ensures fairness and consistency in the evaluation process.

The resulting precision (0.978), recall (0.998), and F1 score (0.9877) in Table 3 highlight the LLM’s exceptional ability to accurately identify medications in this data set, regardless of whether they were extracted as combined entities or separate components. A detailed discussion of possible factors contributing to this performance is provided in Section 6.

### 5.2 Application of results to Process Mining

During the validation process, our expert physician grouped all medications into categories. These categories were based on medication classifications like SNOMED CT and typical treatments for COPD. This categorization was essential for filtering out non-COPD medications and simplifying the PM model (see Table 4).

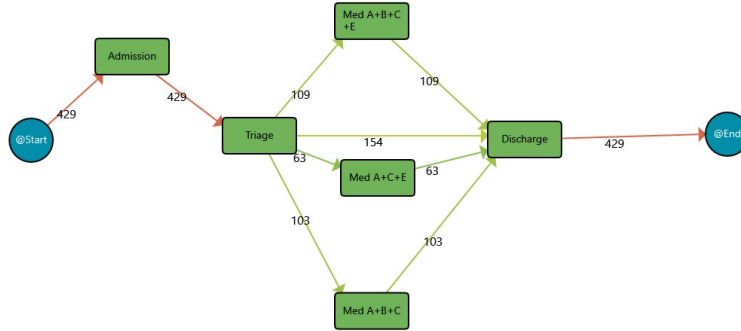
The extracted medications were distributed across **6 categories**. The most common category was **bronchodilators**, which accounted for **1379 instances**. This category included only **8 unique medications**.

We developed a **PM model** to analyze the treatments administered to patients. The model included the extracted medications as events, grouped into categories for easier representation (see Figure 2). The model was designed to represent treatment sequences as events. By analyzing it, we were able to identify patterns and variations in medication administration during emergency visits: for example, antibiotics (E) and inhaled glucocorticoids (B) were more commonly administered to patients with repeat emergency visits, whereas the use of

**Table 4.** Medication categories as designed by a physician and some notable medications

| Category <sup>1</sup>                | Notable Medications                     |
|--------------------------------------|-----------------------------------------|
| A. Bronchodilators                   | Ipratropium bromide, Salbutamol         |
| B. Inhaled Glucocorticoids           | Budesonide, Beclomethasone, Fluticasone |
| C. Systemic Glucocorticoids          | Methylprednisolone, Hydrocortisone      |
| D. Anti-inflammatory PDE4 Inhibitors | Roflumilast                             |
| E. Antibiotics                       | Levofloxacin, Ceftriaxone               |
| F. Mucolytics                        | Acetylcysteine, Mesna                   |
| N. Non-COPD                          | Paracetamol, Furosemide, Omeprazole     |

inhaled glucocorticoids was less frequent within single-visit traces. These findings could inform future care protocols and highlight areas for improvement.

**Fig. 2.** PM model focused on patient treatment, including events for medication administration during emergency care in each COPD-related category, with an applied abstraction of 50%.

Each event in the PM model corresponds to one or more medications administered during a specific emergency visit. These events were defined based on the top-level categories of medications (e.g., “A. Bronchodilators,” “B. Inhaled Glucocorticoids,” etc.). For example, the administration of a bronchodilator was represented as a “Med A” event, while the co-administration of a bronchodilator and an inhaled glucocorticoid during the same visit was represented as a “Med A+B” event. This approach allowed us to capture combinations of treatments while maintaining a manageable level of detail.

## 6 Discussion and Conclusions

This study demonstrates the practical feasibility of using general-purpose LLMs, specifically Llama 3 [11], for extracting medication names from Spanish-language EHR notes. Through an accessible, prompt-based approach, we successfully identified 3739 medication mentions across 122 unique medications, enabling the creation of meaningful process mining (PM) models—something that would otherwise be challenging given the unstructured nature of clinical text.

However, the high performance metrics observed in Section 5.1 could be partially attributed to the characteristics of the notes selected from our dataset (see Section 3.2). These notes typically contained fairly concise and clear information, which makes them easier for the LLM to process accurately. Future studies could explore the performance of LLMs on more complex datasets to evaluate their robustness in handling diverse clinical contexts.

A known limitation of LLM-based extraction is its limited contextual understanding. Medications may appear in clinical notes in ways that do not imply actual administration, such as mentions of contraindications, allergies, or drugs not prescribed. While more advanced methods exist to address these nuances [9], they often require annotated datasets, fine-tuning, or complex pipelines—requirements that conflict with the accessibility goals of our approach. Practitioners must carefully weigh these limitations against the specific requirements of their use cases.

Despite these challenges, our findings highlight the practical benefits of general-purpose LLMs as a first step toward unlocking unstructured clinical data. This lightweight method lowers technical barriers and enables rapid exploratory analysis. For applications prioritizing speed and ease of use—such as hypothesis generation, exploratory PM, or prototyping—our approach can deliver immediate value. However, tasks requiring high accuracy or clinical reliability may find better results investing in more robust NLP workflows.

Future work could explore how this LLM-based method compares to more advanced NLP solutions across diverse datasets, quantifying trade-offs in accuracy, complexity, cost, and scalability. Such research would enable healthcare professionals to make informed decisions about the most appropriate approach for their specific contexts and constraints.

Additionally, future studies could investigate prompts designed to correct misspelled medication names or normalize brand names into their generic equivalents. These strategies introduce additional complexity by combining detection and normalization into a single step, requiring careful evaluation of their performance. Proper assessment would involve separating these tasks (e.g., via a sequential LLM pipeline) or designing unified prompts with clear metrics for each task. While such methods could reduce human post-processing, their increased complexity must be justified by measurable improvements in efficiency or accuracy.

In conclusion, this paper demonstrates how prompt-based extraction techniques can support clinical PM without requiring extensive NLP expertise or annotated reference data. We illustrate this in a real context and provide a easy-

to-use tool to assist experts in classifying LLM results. Furthermore, the tool allows the experts to endorse their identification with SNOMED-CT.

By illustrating the practical steps, challenges, and achievable outcomes, we aim to equip healthcare IT practitioners with a simple yet effective approach to leveraging unstructured clinical data for process mining and other exploratory applications.

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